

Forensic Examinations



CORNCON 12: Tuan Phan
THE 12 CHAOS MONKEYS OF CYBER
CornCon Cybersecurity Conference
Davenport, IA • October 1-3, 2026

Into Insider Threats

About Tuan (Phanrensic)

AVP, eDiscovery and
Digital Forensics.

EF/DF/DFIR, Data
Security & Insider
Threat Strategy

Senior Engineer

ThreatReel

Podcast

<https://threatreel.com>



Slide Deck & Contact



<https://github.com/phanrensic/slides>



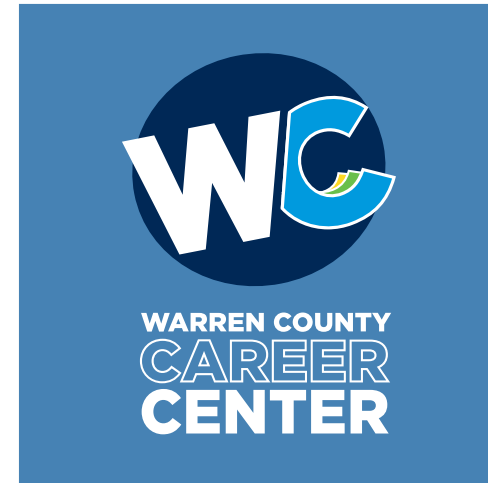
<https://www.linkedin.com/in/tuanqphan>

Volunteer Work

Technical Mentor



Advisory Board



Disclaimer!

Yes, I have a day job.
However...

Opinions expressed are based solely on my own independent security research and do not express or reflect the views or opinions of my employer.

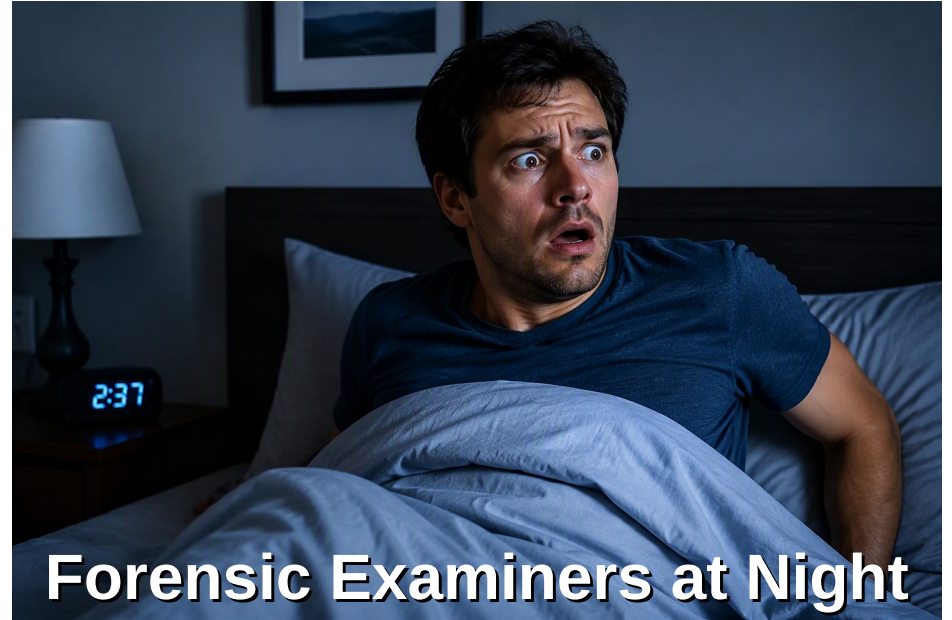


Agenda

Motivations, Risks, Stories

Detect & Investigate

- Lockout Bypasses
- Geolocation Spoofing
- Unauthorized Access



Forensic Examiners at Night

Insider Threat Motivations

- Unapproved Travel
- Over-employment Schemes
- Unauthorized Contracting
- Nation-state Agents & Operatives (Mules)

Organizational Risks

- Financial Loss
- Regulatory Penalties
- Operational Downtime
- Erosion of Trust

True Stories

- Wells Fargo
- DPRK Fraudulent IT Workers

True Story: Wells Fargo

Dozen+ Employees
Fired

- Location?

Simulating Keyboard
Activity



<https://www.cbsnews.com/news/wells-fargo-fires-employees-faking-work/>

True Stories: DPRK

DPRK (North Korean)

Fraudulent IT Workers

- Agents
- Mules



- Stolen Identities

<https://www.justice.gov/opa/pr/justice-department-announces-coordinated-nationwide-actions-combat-north-korean-remote>

Account Lockout Bypasses

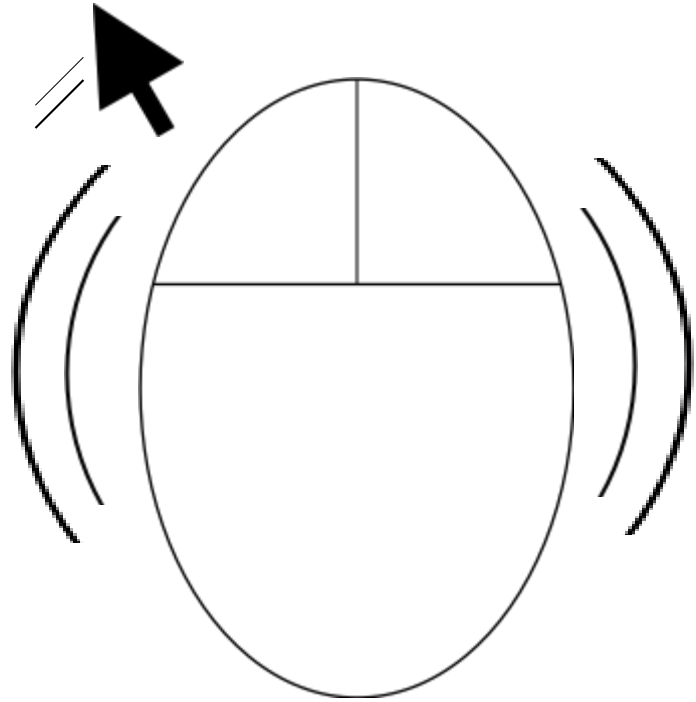
- Techniques
 - Mouse Jigglers
 - Keyboard Key Toggliers

Additional Risks

- Policy Violations
- Unattended & Unlocked Systems
- Stolen Productivity & Integrity
- Potentially Hides Trojan'ed Malware

Mouse Jigglers

- Hardware & Software
- Also called a
 - Mouse Mover
 - Mouse Buddy



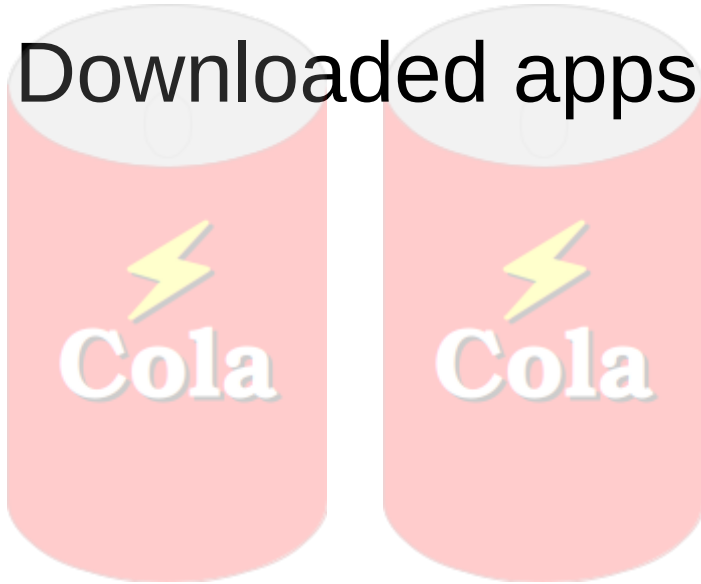
Hardware Devices

- Online Stores
 - USB dongles (HID)
 - Mechanical Platforms



Software Jigglers & Togggers

- caffeine.exe
- Downloaded apps

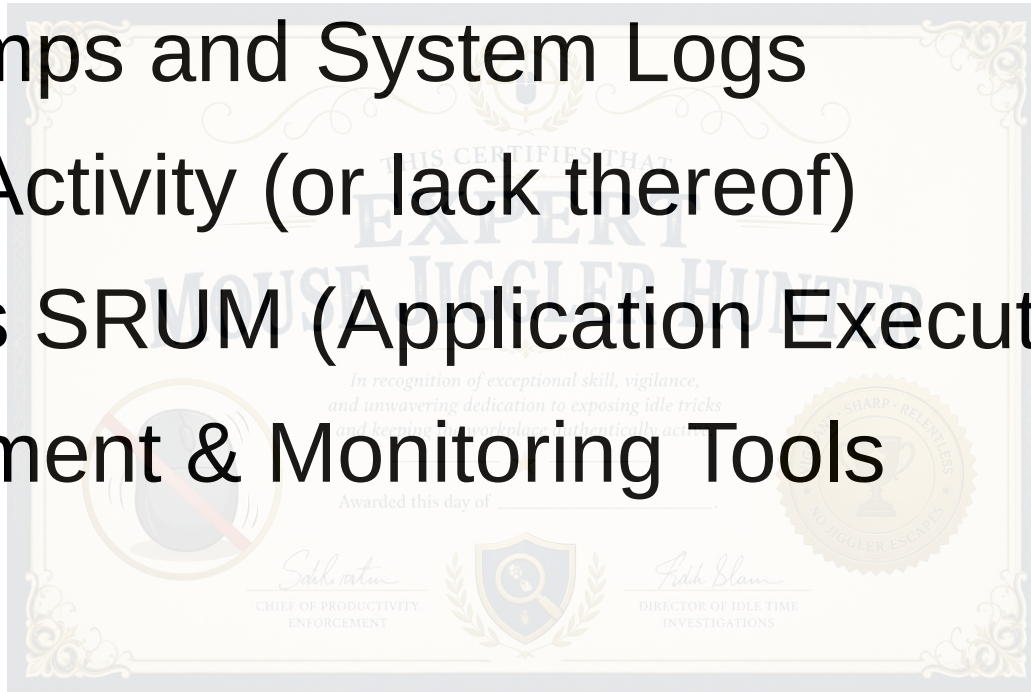


- Custom scripts, macros, etc.

```
min = 5
loop {
    sleep, % min*60*1000
    mousemove, 10, 0, 20, R
    mousemove, -10, 0, 20, R
}
```

Behavioral Detections

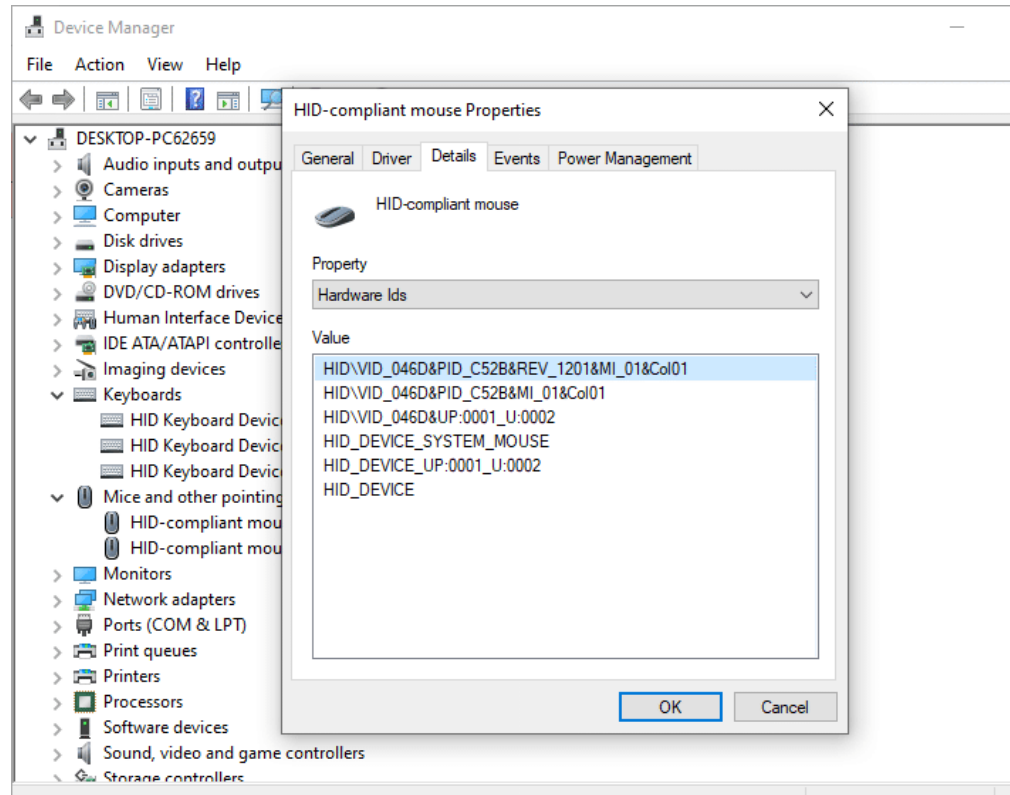
- Timestamps and System Logs
- System Activity (or lack thereof)
- Windows SRUM (Application Execution)
- Management & Monitoring Tools



Device Signatures

- Vendor ID (VID): 0x0E90
- Product ID (PID): 0x0028 or 0x0045
- USB device class
 - Windows Registry, USBSTOR, syslog

Windows Artifact Example



Further Reading

"How to Detect Mouse Jiggler Activity in the Workplace"

By: Jahnavi Pulluri



<https://www.timechamp.io/blogs/how-to-detect-mouse-jiggler-activity-in-the-workplace/>

Geolocation Spoofing



Laptop Farms



Connectivity Artifacts

**VPN
Logs**

**Proxy
Logs**



SSID



BSSID



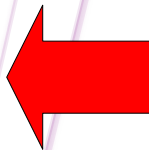
PCAP

101010
101010
101010

VPN Logs

log_timestamp	user_dst	city_source	state_source	country_source	postal_source	longitude_source	latitude_source	event_description	ip_source
2026-01-01T17:38:09.000-06:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Closed connection	passed_hc_policies:['OS_Check\$\$ Asset Certificate']
2026-01-14T13:30:38.000-06:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Closed connection	passed_hc_policies:[' Asset Certificate\$\$OS_Check']
2026-03-02T19:32:44.000-06:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Closed connection	75.66.207.3
2026-03-10T17:29:30.000-05:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Agent login succeeded	75.66.207.3
2026-04-09T10:28:47.000-05:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Agent login succeeded	primary auth-server name: [2factor_auth]
2026-04-12T11:33:02.000-05:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Agent login succeeded	primary auth-server name: [2factor_auth]
2026-04-26T15:42:26.000-05:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Closed connection	75.66.207.3
2026-04-27T10:23:47.000-05:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Agent login succeeded	primary auth-server name: [2factor_auth]
2026-04-29T10:31:31.000-05:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Agent login succeeded	primary auth-server name: [2factor_auth]
2026-05-26T15:04:19.000-05:00	JDOE999	Memphis	Tennessee	United States	38122	-89.9274	35.1567	Closed connection	75.66.207.3

Proximity & Proxy Logs

- Localized Web Traffic
 - Example: google.com vs **google.co.??** 
- Hotel Wi-Fi or other “Captive Portal” locations

Wi-Fi Profile

```
netsh wlan show profile
```

```
Microsoft Windows [Version 10.0.18363.836]  
(c) 2019 Microsoft Corporation. All rights reserved.
```

```
C:\Users\om>netsh wlan show profile
```

```
Profiles on interface Wi-Fi:
```

```
Group policy profiles (read only)
```

```
-----  
<None>
```

```
User profiles
```

```
-----  
All User Profile      : Moto G Play 1087 2  
All User Profile      : vivo Y31L  
All User Profile      : vivo 1802  
All User Profile      : FLGM-bWVnaGFrYWtrYXI5MDU4  
All User Profile      : Moto G Play 1087  
All User Profile      : LPU Hostels  
All User Profile      : JioNet@LPU  
All User Profile      : LPU Wireless  
All User Profile      : Den Boomband
```

```
C:\Users\om>
```

Wi-Fi Interface

```
netsh wlan show interface
```

There is 1 interface on the system:

Name	: Wi-Fi
Description	: Intel(R) Dual Band Wireless-AC 3165
GUID	: d0f2b8e0-66a5-4b6d-bc3c-e3b30baa95fd
Physical address	: 00:e1:8c:31:4d:cc
State	: connected
SSID	: My-Network
BSSID	: fc:dd:55:67:bb:b9
Network type	: Infrastructure
Radio type	: 802.11n
Authentication	: WPA2-Personal
Cipher	: CCMP
Connection mode	: Profile
Channel	: 5
Receive rate (Mbps)	: 150
Transmit rate (Mbps)	: 150
Signal	: 99%
Profile	: My-Network
Hosted network status	: Not available

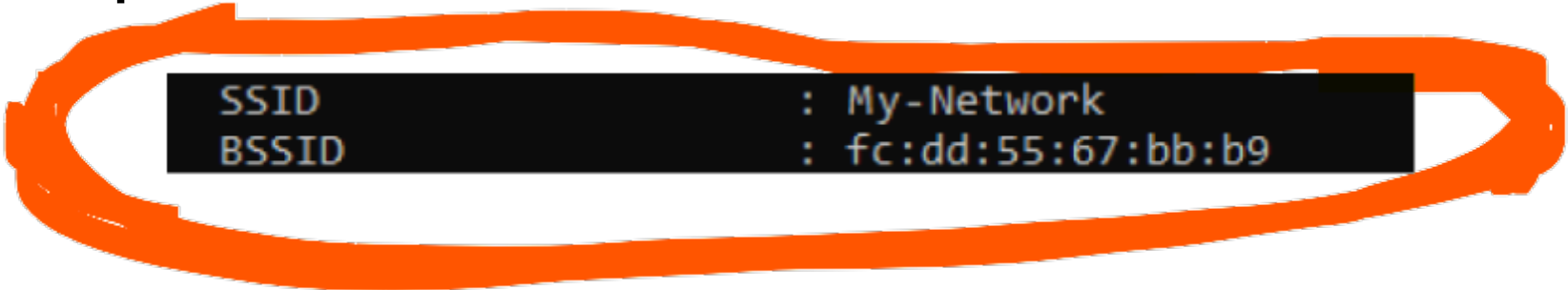
SSID & BSSID

Service Set Identifier (SSID)

- Wireless Access Point (AP) Name

Basic Service Set Identifier (BSSID)

- Unique AP MAC Address



```
SSID      : My-Network  
BSSID     : fc:dd:55:67:bb:b9
```

“wigle.net” Lookup

The screenshot shows the Wigle.net website interface. The browser address bar displays the URL: <https://wigle.net/mapsearch?maplat=51.50830958528049&maplon=-0.1321518211>. The website has a navigation bar with icons for View, Uploads, Info, Stats, and Tools. A search bar is labeled "search for networks". Below the search bar, there are tabs for WIFI, Cell, and BT. The search results show a network named "UnileverWiFi" with QoS: 7, type: infra, and BSSID/MAC: B4:5D:50:AA:86:41. The network was last updated on 20010925174546. The SSID / Network Name is "foobar or foobar%". There are three checkboxes: "Only Free Nets", "Only Commercial/Pay Nets", and "Only Nets I Was the First to See". On the right side, there is a map of London with a yellow pin indicating the location of the network. The map shows various districts including Islington, Hackney, Bethnal Green, City of London, Westminster, Lambeth, Bermondsey, Peckham, Camberwell, Vauxhall, Chelsea, Fulham, Amersmith, Shepherd's Bush, Kensington, Marylebone, and Chelsea. The map is labeled "WIGLE.NET" and "London".

search for networks

WIFI Cell BT

Lat: 51.4051 to: 51.6113

Lon: -0.2969 to: 0.0326

Last Updated: 20010925174546

BSSID/MAC: B4:5D:50:AA:86:41

SSID / Network Name (wildcards: % and _):

foobar or foobar%

☐ Only Free Nets

☐ Only Commercial/Pay Nets

☐ Only Nets I Was the First to See

Search Address

Map

“whereisbssid.py” Lookup

<https://github.com/Trackbool/WhereIsBSSID/>



```

[!] The BSSID Geolocation is based in a database with about 3400000 BSSIDs
[!] Some BSSIDs may not be on the database

[-] BSSID --> 78:8D:F7:CB:A3:F8
[-] Show Location
Sending request...
[#] Extracted Data:
[*] Location: "Carrer Castellvell, 10, 08905 L'Hospitalet del Llobregat, Spain"
[*] Latitude: 41.37094966836
[*] Range: 64.662
[*] Longitude: 2.10804722267
[*] Time: 1499364537

root@deftest:~/Desktop/BSSIDLocation#

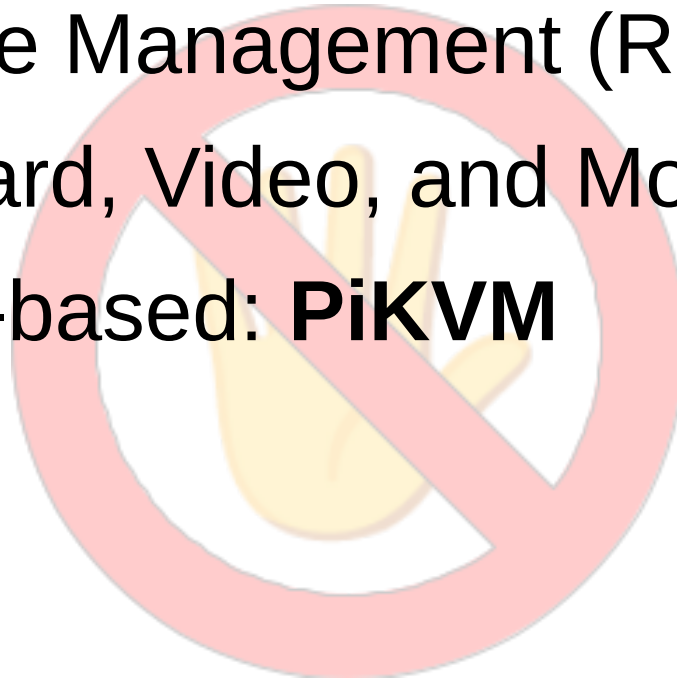
```

Unauthorized Access

Remote Machine Management (RMM) Apps

Network Keyboard, Video, and Mouse (IP-KVM)

- Raspberry Pi-based: **PiKVM**



Hunting RMM Apps

“RMM Threat Intelligence
Tracker”

By: Threat-Hunting-Journal



<https://github.com/Threat-Hunting-Journal/RMM-Tool-Tracking>

The RMM that's Built-In

“We Need to Talk
About Microsoft
Quick Assist”

By: Tyler Hudak



<https://inversion6.com/insights/blog/microsoft-quick-assist-an-it-security-primer>

PiKVM

- Network connected & USB
- 1 device, multiple hosts
- ~\$300 USD



PiKVM Security Risks

- Internet Accessibility
- Works like any other USB peripheral
- Easily abused for unauthorized access

Network Detection

DNS, Proxy, and SSH

- Connection strings with “pikvm” as a username
- Potential mismatches of SSH host keys in the “**known_hosts**” file

Other Strategies

EDR telemetry

MAC address vendor OUI matches

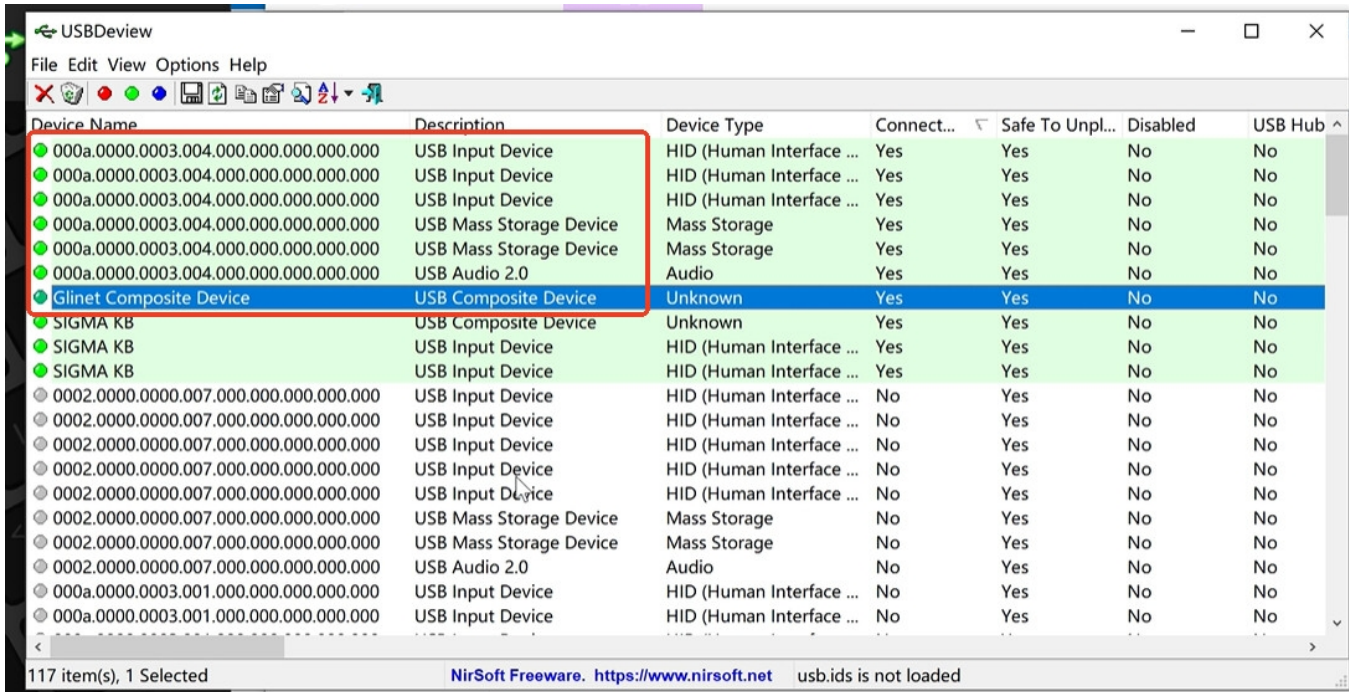
- Sispeed, Raspberry Pi Foundation, GL.iNET

USB Detections

- Hardware IDs or descriptors
 - Particularly generic USB device names (KVM?)
- **lsusb** (Linux)
- Device Manager (Windows)



USB Example



USBDeView

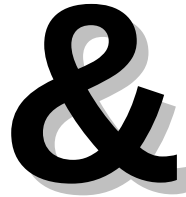
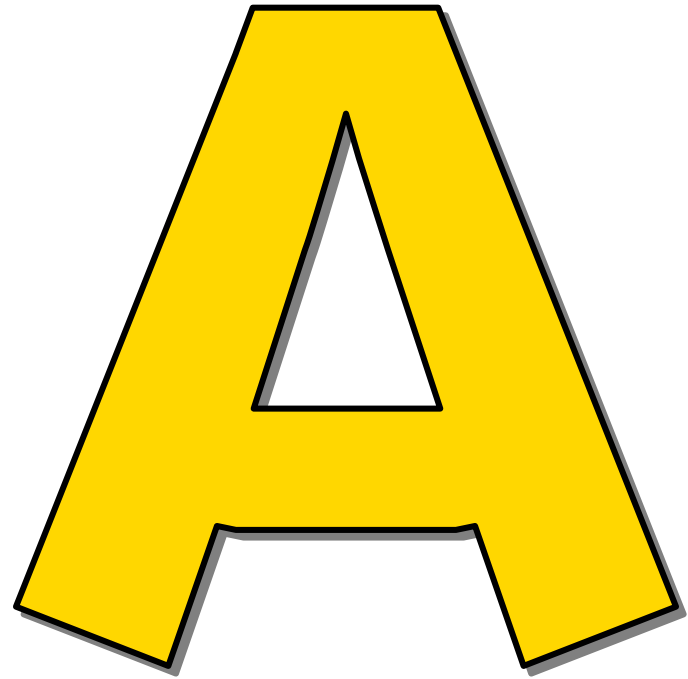
File Edit View Options Help

Device Name	Description	Device Type	Connect...	Safe To Unpl...	Disabled	USB Hub
000a.0000.0003.004.000.000.000.000.000	USB Input Device	HID (Human Interface ...	Yes	Yes	No	No
000a.0000.0003.004.000.000.000.000.000	USB Input Device	HID (Human Interface ...	Yes	Yes	No	No
000a.0000.0003.004.000.000.000.000.000	USB Input Device	HID (Human Interface ...	Yes	Yes	No	No
000a.0000.0003.004.000.000.000.000.000	USB Mass Storage Device	Mass Storage	Yes	Yes	No	No
000a.0000.0003.004.000.000.000.000.000	USB Mass Storage Device	Mass Storage	Yes	Yes	No	No
000a.0000.0003.004.000.000.000.000.000	USB Audio 2.0	Audio	Yes	Yes	No	No
Glinet Composite Device	USB Composite Device	Unknown	Yes	Yes	No	No
SIGMA KB	USB Composite Device	Unknown	Yes	Yes	No	No
SIGMA KB	USB Input Device	HID (Human Interface ...	Yes	Yes	No	No
SIGMA KB	USB Input Device	HID (Human Interface ...	Yes	Yes	No	No
0002.0000.0000.007.000.000.000.000.000	USB Input Device	HID (Human Interface ...	No	Yes	No	No
0002.0000.0000.007.000.000.000.000.000	USB Input Device	HID (Human Interface ...	No	Yes	No	No
0002.0000.0000.007.000.000.000.000.000	USB Input Device	HID (Human Interface ...	No	Yes	No	No
0002.0000.0000.007.000.000.000.000.000	USB Input Device	HID (Human Interface ...	No	Yes	No	No
0002.0000.0000.007.000.000.000.000.000	USB Input Device	HID (Human Interface ...	No	Yes	No	No
0002.0000.0000.007.000.000.000.000.000	USB Mass Storage Device	Mass Storage	No	Yes	No	No
0002.0000.0000.007.000.000.000.000.000	USB Mass Storage Device	Mass Storage	No	Yes	No	No
0002.0000.0000.007.000.000.000.000.000	USB Audio 2.0	Audio	No	Yes	No	No
000a.0000.0003.001.000.000.000.000.000	USB Input Device	HID (Human Interface ...	No	Yes	No	No
000a.0000.0003.001.000.000.000.000.000	USB Input Device	HID (Human Interface ...	No	Yes	No	No

117 item(s), 1 Selected

NirSoft Freeware. <https://www.nirsoft.net> usb.ids is not loaded

Questions?

A large, bold, green capital letter 'Q' with a black outline and a slight drop shadow.A black ampersand (&) with a black outline and a slight drop shadow.A large, bold, yellow capital letter 'A' with a black outline and a slight drop shadow.

The End



Thank you for
attending
**Forensic Examinations
into Insider Threats!**